Sheet 58 Mon 5 Apr. 2027 *Plane geometry problems: choosing a method*

1. Warm-up · Quick Distance AB for $A(0;0)$ and $B(5;12)$?
2. Core · Method $A(0;0)$, $B(4;2)$, $C(-2;4)$. What type of triangle is ABC?
3. Core + · Method Give two methods to prove, using coordinates, that $ABCD$ is a parallelogram.
4. Review · Reading In a class of 30 pupils, a bar chart shows x pupils with the mark 12 and $2x$ pupils with the mark 14; the other marks account for 18 pupils. Find x.
5. Thread · Quick A pyramid has a square base of side 6 cm and height 5 cm. Calculate its volume.

Sheet 59 Tue 6 Apr. 2027 *Situations from other subjects*

1. Warm-up · Quick A moving object has position $x(t) = 3t + 2$ (in m, t in s). Where is it at $t = 4$?
2. Core · Method A second object follows $x_2(t) = 5t - 4$. At what time and where do the objects meet?
3. Core + · Quick A velocity vector has coordinates $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ (in m/s). What is its magnitude?
4. Review · Quick Calculate $\dfrac{1}{2} - \dfrac{3}{5}$.
5. Thread · Write-up ABC is isosceles at A and H is the foot of the altitude from A. What can be stated about H?

Sheet 60 Thu 8 Apr. 2027 *Communicating a proof: implication, converse, equivalence*

1. Warm-up · Quick State the converse of: "if a quadrilateral is a rhombus, then its diagonals are perpendicular".
2. Core · True/false True or false? Justify. The previous converse is true.
3. Core + · Write-up Is the equivalence "$x^2 = 4 \Leftrightarrow x = 2$" correct? Correct it.
4. Review · Quick Compare 2^{10} and 10^3.
5. Thread · Quick Between which consecutive integers does $\dfrac{7}{4}$ lie on a number line? Give its decimal value.

Sheet 61 Mon 26 Apr. 2027 *Class discussion: comparing methods*

1. Warm-up · Quick Distance between $A(-2;1)$ and $B(1;5)$?
2. Core · Method Give three methods to prove that A, B, C are collinear.
3. Core + · Reflection To prove that two lines given by $y = m_1x + p_1$ and $y = m_2x + p_2$ are parallel, which method is quickest?
4. Review · Factorise Factorise $6x + 9$.
5. Thread · Method Write 360 as a product of prime factors. How many divisors does it have?

Synthesis problems, reasoning and starter routines

Sheet 62 Tue 27 Apr. 2027 *Synthesis problems across several areas of the curriculum*

1. Warm-up · Quick Solve $5x - 3 = 2x + 9$.
2. Core · Problem A price rises by 20 %, then a fixed discount of 1 000 Ar is applied, giving 11 000 Ar. What was the original price?
3. Core + · Method $f(x) = 2x + 1$ and $g(x) = -x + 7$. Find the point of intersection of the curves, then solve $f(x) < g(x)$.
4. Review · Quick A recipe for 4 people needs 250 g of flour. How much for 6 people?
5. Thread · Quick A shape of area 4 cm^2 is enlarged with scale factor 3. What is the area of the enlargement?